



Engr. Abdul Wahab

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RESEARCH INTERESTS

AI Mobility Systems

Autonomous and Assisted Vehicles

UAV (Unmanned Aerial Vehicle)

Connected Autonomous Vehicles (CAVs)

Vehicle Design and Engineering

Autonomous Vehicle Systems

Vehicle Dynamics and Control

PROFILE SUMMARY

My name is Abdul Wahab, and I am actively seeking an opportunity and environment where I can implement my expertise, my skills, my knowledge, and my foundation to solve the current problems. I am highly interested to learn and gain hands-on experience in above mentioned research areas because, in the coming years, tremendous advancements will happen in AI mobility. As a mechanical engineer with skills and knowledge in Autonomous systems, this direction is the right choice for me to go and become a valuable resource and contribute meaningfully to the benefits of humanity.

WORK EXPERIENCE

June 2026 — September 2026

Mechanical Design Engineer

[Dysol \(Pvt\) Ltd, Lahore, Pakistan](#)

August 2025 — Present

Mechanical Design Engineer

[Upwork \(National & International Clients\)](#)

June 2026 — Present

Web & Application Developer — Applied AI Enthusiast

[Self Employed](#)

EDUCATION & TRAINING

September 2019 — October 2023

B.Sc. Mechanical Engineering

[University of Engineering and Technology \(UET\), Peshawar — KPK, Pakistan](#)

FINAL YEAR THESIS PROJECT

Design and Fabrication of Mechanism for Automated Installation of 18-Liter Water Bottle for Water Dispensers

Supervised by **Prof. Usman**

Thermodynamics

Fluid Mechanics

Machine Design

CAD / CAM

Manufacturing Processes

Heat Transfer

Mechanics of Materials

Engineering Dynamics

TECHNICAL SKILLS

Mechanical Design & Engineering Simulation

- 3D CAD Modeling & Drafting (SolidWorks)
- Design for Manufacturing and Assembly (DFMA)

- Kinematics & Mechanical Systems Integration
- Finite Element Analysis (FEA) & CFD (ANSYS)
- Kinematic & Dynamic Modeling (Altair EDEM)

Autonomous Systems & Software Integration

- Advanced Driver Assistance Systems (ADAS) & Collision Avoidance
- Path-Tracking, Control Algorithms & Vehicle Dynamics Modeling
- Programming (Python, its libraries) & Simulation Environments
- Application Development (Streamlit) & Applied AI (LLMs)

COURSES & CERTIFICATIONS

- ✦ **IELTS Academic** — Band 6.0 (Listening, Reading, Writing, Speaking)
- ✦ **Certified SOLIDWORKS Associate (CSWA)** — Dassault Systèmes
- ✦ **Pakistan Engineering Council (PEC)** — Registered Engineer
- ✦ **Python & Machine Learning** — codanics.com
- ✦ **ABC of Statistics** — codanics.com
- ✦ **AI ka Chilla 2026** — codanics.com

Additional Courses

- ✦ **Lean Six Sigma Yellow Belt (LSSYB)** — udemy.com
- ✦ **Lean Six Sigma White Belt (LSSWB)** — udemy.com
- ✦ **Introduction to Minitab** — simplilearn.com
- ✦ **CPR, AED & FIRST AID** — alison.com

FEATURED PROJECTS

1. Fitness Product Development — Mechanical Dispensing System

Industry: Fitness & Consumer Products

Challenges Managed:

Optimizing complex mechanical interactions while maintaining compact design requirements. Achieved functional mechanism performance through iterative CAD refinement and simulation.

Tools Used:

SolidWorks (3D Modeling, Assembly Design, Motion Analysis), Altair EDEM (Simulation)

2. Medical Device Development — Controlled Flow Mechanical System

Industry: Medical Technology

Challenges Managed:

Maintaining reliable mechanical operation under strict performance requirements. Improved design reliability through engineering analysis and simulation.

Tools Used:

SolidWorks (CAD Modeling & Motion Analysis), ANSYS (Simulation & Validation)

AI FEATURED PROJECTS

Doc Agent — AI Research Assistant

[View Live App ↗](#)

An intelligent document analysis system built to make Research & Development incredibly easy. Users can upload any type of file and interactively ask questions to extract complex knowledge and information hidden within the documents instantly.

Python

Streamlit

Large Language Models (LLMs)

Document Parsing

2D Autonomous Vehicle Tracker

[View GitHub ↗](#)

Developed a 2D simulation environment using Python and Pygame to model and evaluate autonomous vehicle control algorithms. Implemented a Kinematic Bicycle Model with Euler forward integration to simulate realistic vehicle physics. Engineered a Pure Pursuit Path-Tracking Controller with dynamic look-ahead distances for smooth geometric trajectory following along complex S-curves, visualized through a real-time telemetry dashboard.

Python

NumPy

Pygame

Kinematics

Control Algorithms

Adaptive Cruise Control (ACC) & AEB Simulator

[View GitHub ↗](#)

Built a dynamic longitudinal control simulator for an Adaptive Cruise Control (ACC) and Autonomous Emergency Braking (AEB) system. The control algorithm calculates the desired following gap dynamically based on velocity, and utilizes simulated radar perception with Gaussian noise to mimic real-world sensors. Calculates real-time Time-To-Collision (TTC) to engage emergency braking when a collision is imminent, ensuring vehicle safety against a dynamic lead vehicle.

Python

ADAS

Mathematical Modeling

Pygame

Collision Avoidance

LANGUAGES

Language	Level
English	Proficient (B2)
Urdu	Fluent (C2)
Seraiki	Native (C2)

ACADEMIC REFERENCES

Prof. Dr. Naveed Ahmad

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Prof. Muhammad Alam Zaib Khan

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